NATIONAL UNIVERSITY OF SINGAPORE

CS5224 – Cloud Computing

(Semester 2: AY 2025/2026)

Time Allowed: 1.5 Hours

Sample Answers

INSTRUCTIONS TO STUDENTS

1. This assessment paper contains **TWENTY FIVE (25)** questions.
2. Total marks in this question paper is **30**.
3. Please answer **ALL** questions.
4. This is a **NON-SECURE BLOCK INTERNET** assessment.
5. This is an **OPEN BOOK** assessment. You are allowed to get reading materials as well as use study materials on your laptop.
6. You are not allowed to use any AI tools in this assessment and this includes the use of any locally installed Large Language Models (LLMs).

Examiner's Use Only

Question No.	Maximum Marks
MCQs 1 – 20	20
21	2
22	2
23	2
24	2
25	2
Total	30

Question 1

How does an Application Load Balancer enable reliability?

(a) By routing traffic to the least busy instances
(b) By routing traffic away from failed instances
(c) By caching frequently accessed content
(d) By replacing failed instances
(e) By routing traffic to new instances

Option (b).

Question 2

You wish to build your own application and are operating in a highly competitive market where being first is more important than being perfect, and rapid iteration is critical. Which cloud service model is most appropriate?

(a) IaaS
(b) FaaS
(c) CaaS
(d) PaaS
(e) SaaS

Option (d). PaaS enables the fastest building and deployment. IaaS takes time while SaaS is too restrictive since you wish to build your own application here.

Question 3

Which one of the following statements is correct regarding virtualization?

(a) Virtualization increases compatibility among hardware that use different standards
(b) Operating System virtualization is better suited for an application which requires direct communication with hardware resources
(c) In hardware-assisted virtualization, the guest operating system directly communicates with hardware resources
(d) Virtualization technology allows the isolation of applications running on the same server
(e) Virtualization allows multitenancy only when it is Type 2

Option (d) – comes directly from the definition in some ways

Question 4

A company wants to use MySQL as the database for its web application while selecting the PaaS cloud service model. Which of the following is the most appropriate?

(a) The company can use AWS RDS (Relational Database Service) and configure a MySQL database
(b) The company needs to install a MySQL server
(c) The company should use MongoDB instead of a MySQL server
(d) The company should use the LAMP stack, which contains Linux, Apache web server, MySQL, and Python
(e) It is not possible to use MySQL with PaaS

Option (a). For a PaaS, you configure MySQL. LAMP is an IaaS stack.

Question 5

A public-sector platform must keep both data and metadata within one country's borders and prevent foreign access under all circumstances. Which cloud deployment model is the best fit?

(a) Public Cloud
(b) Private Cloud
(c) Hybrid Cloud
(d) Sovereign Cloud
(e) Multi Cloud

Option (d). Answer follows directly from bookwork

Question 6

A retailer normally runs their infrastructure on-premise but wants excess holiday traffic to spill over into pre-deployed cloud resources only when local thresholds are exceeded, then return to on-premise when the load drops. Which architecture best matches this?

(a) Resource pooling architecture`
(b) Service load balancing architecture
(c) Cloud bursting architecture
(d) Elastic resource capacity architecture
(e) Using horizontal or vertical scaling architecture

Option (c). Answer follows directly from bookwork

Question 7

Which requirement is the clearest sign that a workload is a poor fit for FaaS?

(a) It is triggered by file uploads
(b) It requires long-lived stateful sessions and persistent in-process connections
(c) It performs short request-driven transformations
(d) It runs sporadically and scales by event volume
(e) It requires long-lived stateless sessions

Option (b). Rest all work fine for FaaS

Question 8

A company has already made significant investments in on-premise infrastructure. Migrating to the cloud would require substantial additional costs for integration. Which of the following is the most appropriate conclusion based on this scenario?

(a) Cloud is always cheaper because it reduces upfront capital expenditure
(b) Reserved cloud pricing eliminates migration and integration costs
(c) On-premise solutions always have lower operational costs than cloud
(d) Past investments should not be considered when evaluating future decisions
(e) Cloud adoption may be less attractive due to high migration and integration costs

Option (e). (a) ignores sunk costs – may be cheaper later, but no guarantee. (b) and (c) are not really options. (d) can't ignore sunk costs

Question 9

"Great power and control come with greater responsibility."

Which cloud service model best reflects this statement?

(a) SaaS
(b) FaaS
(c) CaaS
(d) IaaS
(e) PaaS

Option (d). (IaaS) provides the highest level of control over computing resources (VMs, storage, networking). However, this also means the user is responsible for managing the OS, updates, scaling, and security.

Question 10

Which of the following features is the **primary distinguishing factor** between Virtual Machines (VMs) and Docker containers?

(a) Isolation
(b) Security
(c) Hardware virtualization
(d) Command-line Interfaces
(e) Separate namespaces

Option (e). Virtual Machines rely on hardware virtualization, which introduces additional overhead. Docker containers, on the other hand, share the host OS kernel and avoid this overhead, making them lighter weight.

Question 11

You are tasked with designing a datacenter. Your manager asks you to prioritize components based on increasing energy consumption (from lowest to highest).

Which of the following orders is correct?

(a) Networking → Disk → DRAM
(b) CPU → Disk → Networking
(c) Disk → DRAM → Cooling
(d) Cooling → Networking → DRAM
(e) Disk → Cooling → CPU

Option (e). Based on the lecture's energy breakdown, disks consume the least energy among the listed components, followed by cooling infrastructure, while CPUs typically account for the highest energy consumption.

Question 12

Which of the following best describes a serverless application?

(a) A serverless application is usually developed in the context of PaaS
(b) The developer of a serverless application needs to pay a small fee for the idle periods
(c) A serverless application does not need a server/host to run on
(d) The developer of a serverless application does not need to provision or manage any servers
(e) The developer of a serverless application has lower costs

Option (d).

Question 13

A company manages development, staging, and production environments using Terraform. Which one of the following approaches best avoids duplication and ensures consistency?

(a) Copy-paste configurations per environment
(b) Use separate state files
(c) Use only one environment
(d) Hardcode environment values
(e) Keep separate code repositories

Option (b). Copy-paste can lead to issues while having one environment is prone to error due to its 'monolithic' nature. Hardcoding isn't an option and keeping separate code repositories sometimes makes it harder to manage.

Question 14

Your company has recently gotten to grips with Terraform and wish to use it. However, they also wish to have automatic drift detection during continuous integration (the practice of merging code changes into a central repository frequently). Which of the following is the best approach to achieve this?

(a) Run `terraform apply` daily
(b) Disable drift checks
(c) Use manual audits
(d) Periodically recreate the infrastructure
(e) Run `terraform plan` and provide an alert on changes

Option (e)

Question 15

Which of the following are signs of a highly available application?

(i) Applications are protected behind multiple layers of security
(ii) Spikes in user demand are met through automatically increasing resources
(iii) A failure in one geographic region will trigger an automatic switch to resources in a different region
(iv) Virtualized hypervisor-driven systems are deployed as mandated by company policy
(v) Application developers ensure the necessary software updates are applied

(a) (i) and (iii)
(b) (ii) and (iv)
(c) Only (v)
(d) (ii) and (iii)
(e) (i) and (v)

Option (d). Security and keeping software updated doesn't directly translate into availability of the application.

Question 16

What is the primary challenge when migrating legacy systems to the cloud?

(a) Increased cost
(b) Decreased oversight
(c) Application compatibility
(d) Decreased performance
(e) Increased performance

Option (c). Cloud environment and set-up may be different

Question 17

Your users are reporting increased levels of latency. On checking you notice that CPU usage is actually low. What may be the primary cause then?

(a) Servers are overprovisioned
(b) Too many instances have been set up
(c) Horizontal scaling
(d) Network bottleneck
(e) Usage of CDNs

Option (d). Most probably cause for increase in latency.

Question 18

A company, GreatestCloudOfAll Inc, have noticed that their cloud infrastructure frequently scales up and down, which is causing great instability and hence extra costs. You have been called in as an expert to advise them. What would your advice be?

(a) You just need to increase your number of instances and it'll be fine
(b) You can re-check your thresholds for scaling and also add a cooldown period which prevents further scaling during this period
(c) You can monitor it and over time just disable scaling as that will help in the long run in this case
(d) Best is manual scaling, nothing beats it!
(e) Reduce the overall cost by reducing your number of instances

Option (b). Rest are really not practical. By having a cooldown period, you provide time for resources to adjust before making any changes, as needed.

Question 19

A company wishes to migrate its legacy application to the cloud quickly with minimal changes. Which approach from the following would be most suited (you can assume that all options listed are technically feasible)?

(a) Refactor the application to use serverless functions and then deploy it
(b) Get a cloud provider to build a SaaS equivalent solution for you and then use it
(c) Rewrite the application using a PaaS platform which allows for easy deployment
(d) Use IaaS and move application as is from on-premise to cloud
(e) Keep it on-premise

Option (d). (b) may not be a choice depending on your application. Rest all take much more time and effort. (e) is not really an option.

Question 20

Which one of the following is an optimal strategy to employ when you have a very large number of reads and write operations to a scalable cloud database?

(a) Horizontal scaling
(b) Vertical scaling
(c) Increasing storage
(d) Cloud bursting
(e) Migrate to a faster database

Option (a). Horizontal scaling allows you to distribute these reads and writes across multiple nodes.

Question 21

Assume that you are responsible for organising your cloud resources globally as the leader of an enterprise. Discuss at least **two** key factors that you should consider when deciding on cloud resource hosting.

Answer: Availability, Cost of cloud based on region, Location for latency, Location for legal requirements. This questions is about hosting cloud resources globally and what to look for, and less about models (whether it be deployment or service).

Question 22

Design a cloud strategy for a healthcare analytics SaaS used by multiple hospitals. Patient data must be tightly governed, demand can spike during outbreaks, and the system must still scale nationally. What deployment model(s) and scaling architecture(s) would you choose, and why?

Answer: can use hybrid cloud or private cloud. Can use cloud bursting, or controlled dynamic scaling – scaling should be up and down.

Question 23

You are responsible for the cloud infrastructure in company TheGreatCloud.sg. You have been told that the new system you are designing will be holding sensitive data which must comply with data residency laws across multiple countries. How will you design your data storage to be able to manage this cross-border access while being compliant to the data residency laws? Give **two** ways in which you can achieve this.

Answer: Ensure data in for a region stays within that region and route users to region-specific services, use access controls and encryption, as necessary. Can also add any data auditing, as necessary. Data storage will be fragmented but it has to be compliant. A sovereign cloud is one option but you also need to make sure requests are routed accordingly.

Question 24

A bank wants to move away from its legacy application and modernise it with strict compliance and some sensitive data. Which cloud service and cloud deployment model would you use and why?

Answer: Hybrid-cloud: sensitive workloads on private cloud, and non-sensitive on public cloud. Build your own through PaaS for the modern services while any legacy infrastructure could be on IaaS. Key here is that not all data is sensitive and also that you need to update your application, so build a new one, which implies PaaS.

Question 25

You are a Solutions Architect at a leading technology company. The IT division of the metropolitan MRT (Mass Rapid Transit) system has approached you to design a next-generation payment platform to support SimplyGo and phase out the legacy EZ-Link card system. A key concern is that during peak hours, the current system experiences high latency, resulting in long queues and poor commuter experience. The reason for this is that for each user, the system performs fare calculation, balance deduction, and displays the update immediately before letting the user pass through the gate.

Your task is to propose **two** ways in which this payment platform can be scalable and also have low-latency.

Answer: Primary issue is the sequence of operations before letting a user through. You can do some of these later – just do what is necessary to let user through and other operations can be performed later. Doing these operations async is key and here FaaS or any other similar element can help. As before, scalability must be elastic which allows system to scale up/down depending on user numbers.

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